

Due to broad human individuality, immune imbalance, and dietary options that exist, we have avoided the all too common and false thinking, which recommends a singular diet to unilaterally fit the needs of all patients. Instead, EVND physicians have focused on encouraging dietary recommendations based on precise testing of each individual's metabolic challenges, food sensitivities, and gut microbiome. With the focus on achieving healthy bowel function for the purpose of reducing disease risk, we are guided by the principle of discovering root disturbances and restoring optimal metabolic and gut function. Along the way, we learned and created very helpful guidelines or rules, based on research and experience, which help guide and simplify successful dietary practices for metabolic optimization, reversing diabetes and heart disease, normalizing GI function, and weight loss.

Intestinal Permeability

The intestinal tract may be the most important potential contributor to disease and health. One of the leading concerns that presents in many ill patients, but also puts unsuspecting healthy individuals at risk, is the presence of a “leaky gut”. Although the intestinal tract is contained within the body, it is really an extension of the outside world allowing the introduction of food and water. It serves as the primary physical barrier from the outside world, protecting us from potentially harmful environmental factors passing through the digestive tract, including food additives, pesticides, toxic metals, toxins, pathogenic microorganisms such as candida and parasites, and other potential antigens.

The protective and absorptive intestinal lining consists of a single layer of epithelial cells separated by protein complexes known as tight junctions. In a healthy gut, tight junctions regulate intestinal permeability, selectively allowing the transport of digested proteins, fats, sugars, water, nutrients, and electrolytes across the intestinal barrier, with an equally important priority of preventing the absorption of harmful substances.

Difficulties arise when various environmental factors damage the tight junctions, resulting in increased intestinal permeability often referred to as leaky gut syndrome. Multiple conditions and dietary practices contribute to increased intestinal permeability. Examples include dietary excesses from saturated fat and alcohol, nutrient deficiencies including vitamin D and fiber, antibiotic use, stress, and fungal, bacterial, and parasitic infections.

With the increased intestinal permeability, undesirable antigens are able to cross the gut mucosa into systemic circulation. This results in an immune response resulting in free radicals and chronic inflammation causing fatigue and possible “brain fog.” Leaky gut has also been associated with certain gastrointestinal conditions, such as irritable bowel syndrome, ulcerative colitis, Crohn's disease, celiac disease, type 1 diabetes, and rheumatoid arthritis to name just a recognizable few.

Healing from Intestinal Permeability

The following ideas can be used to address the root imbalances of most digestive conditions while reversing intestinal permeability by supporting gut healing.

1. Remove Overabundant Pathogens Residing in the Intestinal Tract

The first step is to *remove* pathogens including candida, yeast, parasites, and bacteria associated with increased intestinal permeability. Targeting pathogens effectively may require treatment with pharmaceutical medication, antibiotics, or antimicrobial, antiparasitic, and antifungal herbal supplements. Overgrowth of pathogens creates an unhealthy microbiome resulting in fatigue, bloating, brain fog, and many other common health complaints.

2. Reduce Inflammatory Foods and Replace with Healing Foods

The second step is to *replace* inflammatory foods with dietary nutrients and gut-healing foods that help to reduce inflammation and support digestive health.

- Consume good fats from extra virgin olive oil, avocados, and wild-caught fish(Omega-3 fatty acids)
- Consume predominantly high-fiber foods such as vegetables, fruit, and nuts
- Consume anti-inflammatory herbs and spices (e.g., turmeric, rosemary, garlic)
- Avoid saturated, hydrogenated, and trans-fatty acids as found in most processed and fast food choices
- Avoid other food sensitivities(such as gluten and dairy), which may be identified through IgG food intolerance testing or the elimination diet
- Avoid synthetic chemicals such as pesticides, metals, and PCB's in non organic foods
- Avoid processed foods which can include food additives such as flavor enhancers, colors, artificial sweeteners, and MSG
- Avoid refined starches and white refined sugar particularly fructose, dextrose, and high fructose corn syrup (HFCS)
- Avoid alcohol

In addition, digestive enzymes can be added to each meal to support digestion and absorption of dietary nutrients. Digestive enzymes support individuals, with impaired digestive function, by enzymatically digesting fats, proteins, and carbohydrates. This helps reduce the expression of food intolerances such as lactose and gluten intolerance. Other options include bitters or bile acid supplements to assist in nutrient absorption by emulsifying lipids, supporting the function of the liver and gallbladder.

3. Repair the Gut Lining with Advanced Therapies

The final step involves *repairing* the gut lining with specific nutrients, medicinal herbs, and peptide therapies that have been shown to decrease intestinal permeability and inflammation. Examples of these ingredients include:

- Larazotide peptide
- Aloe vera
- Chios mastic gum (*Pistacia lentiscus-Anacardiaceae*)
- Deglycyrrhizinated licorice (DGL)
- L-glutamine
- Marshmallow root (*Althaea officinalis*)

Food Elimination Diets

It can be very challenging to change the diet, but it can bring significant benefits that will not likely come by any other means or treatments. An elimination diet or IgG food intolerance testing are other options to help delineate the right foods to eat and avoid for optimal health. Another good option is the popular Whole 30 Diet designed to eliminate the common inflammatory foods that cause symptoms and illness in so many patients we treat. If you are over the age of 50 and are concerned about brain health as well as cardiovascular health then consider the Keto Flex 12/3 Plus reported to prevent and reverse age-related cognitive decline that also prevents diabetes and heart disease. The 2 most common foods to contribute to inflammation and a variety of symptoms include categorically dairy and either gluten or all grains. Eliminating or greatly reducing dietary intake of grains, gluten, and dairy can contribute to a reduction in fatigue, brain fog, and full body inflammation.

4. Add in Fermented Foods

The consumption of bacterial fermented foods can provide a robust and diverse injection of friendly bacteria to our gut microbiome. This is one of the leading methods for nature to provide health and immunity in the absence of medications or herbal therapies to prevent disease. Many cultures regularly consume fermented foods such as Kefir, Kimchi, Sauerkraut. The at home production of yogurt cultures provides yet another flexible, customizable, and affordable microbiome support therapy for most individuals.

Customized therapy can follow a microbiome stool testing, where a patient's report displays imbalances, deficiencies, and excesses of vital or pathogenic bacteria that make up your individual microbiome. An excess of pathogenic bacteria can be treated with a short course of prescription antibiotics or antibacterial herbal therapies. A deficiency in a family or species of friendly bacteria can be supported with specific oral doses of friendly bacteria called probiotics.

With the abundance of current research revealing multiple strains and the production of so many viable combinations of friendly probiotic strains, it becomes imperative to be as specific as possible to maximize benefit and cost. A customized approach to maximize individual needs and benefit can be to culture or create a customized probiotic combination that is based on our individual microbiome. This is accomplished by growing our own cultures using specific probiotic sources based on diagnoses, symptoms, and our individual microbiome study.

As an example we could take one capsule of the product L. Plantarum PS128 and one capsule of another product called Bravo and when we mix the contents of the individual capsules together in dairy or coconut cream we have manufactured our own customized culture base containing more than 300 different strains. If we then take that base and maintain a set temperature for a specified time we will double the bacterial count of the friendly bacteria creating your own culture. Within a few hours we can have a new probiotic combination that is many times greater than the products with which we started. This new culture can be consumed directly or added to foods, drinks, salads etc; to provide a high dose, multi strain probiotic for a fraction of the cost of dosing a daily single capsule straight from the bottle. We now have personalized therapy with the added benefit of an affordable broad spectrum probiotic.

At home production of yogurt will require a yogurt maker. A good brand and product that holds the temperature well in range for making cultures and utilizes ceramic or glass containers(BPA free) is offered at Luvele.com for around \$110. Luvele provides printed and online instructions for various yogurts including those made with dairy or coconut milk. We provide directions here for production of a non-dairy yogurt utilizing coconut cream.

How to start:

1. To start you will need to purchase a yogurt maker. A good brand and product that holds the temperature well in range for making cultures and utilizes ceramic or glass containers(BPA free) is offered at Luvele.com for around \$110. Luvele provides printed and online instructions for various yogurts including those made with dairy or coconut milk.
2. If you are dairy free, you will need to find preservative and gum free coconut cream products. Any added preservatives in the coconut cream will kill the probiotic cultures you add to it otherwise. One option for tasty yogurt making is the brand Savoy. Some other brands will produce runnier yogurts. Read the ingredients section on the label to choose the best product.
3. Next you will need to purchase the specific strain/products recommended by your doctor based on your individual microbiome test results or match symptoms based on research of specific probiotic strains.
4. The final step is to follow the directions for making a coconut yogurt culture with the products gathered in steps 1-3.

How to make a yogurt culture:

1. Start the culture by adding the recommended or desired probiotic capsules into 2 cans of coconut cream. Pour a small amount of coconut cream into the yogurt maker glass jar.
2. Add the starter probiotic powder bit by bit and gently stir the way you would corn flour. If you do this step too fast or just dump the full contents of the capsule on top of the coconut cream you will just get clumps.
3. Add the rest of the coconut cream and gently mix in. Close the lid and place the bowl into the machine.
4. Start the yogurt maker and leave to ferment for 6 (up to 24) hours at body temperature (no more than 100o F). Once the temperature of the unit has been reached for proper growth the probiotic strains will double every 20 minutes. The longer it cooks the greater the number of strains, but the more bitter it will become. We recommend starting out with a culture made over 2 hrs and 20 minutes. Since the culture doubles every 20 minutes a 10 million culture will turn into over 1.2 billion which is equivalent now to 128 capsules.
5. Once it is done, put it in the fridge for a minimum of 4 hours before serving to allow the yogurt to thicken up.
6. Use in the same fashion as you would use dairy-derived yogurt, or as a base for the recipes below. A typical daily dose should be about one tbsp. Make enough to last about 2 weeks based on the number of family members who may use the culture.